

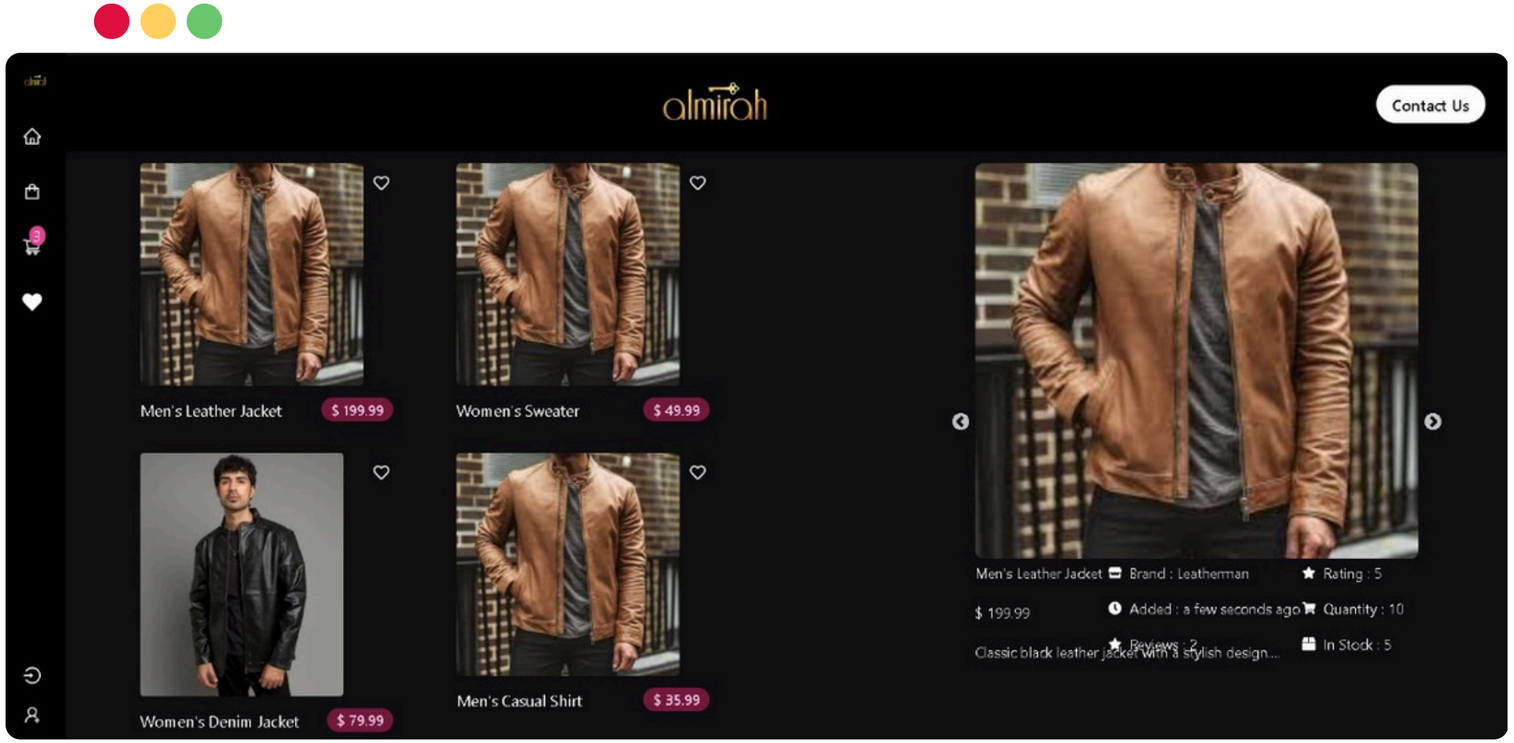
Almirah E-Commerce Platform

Team Members:

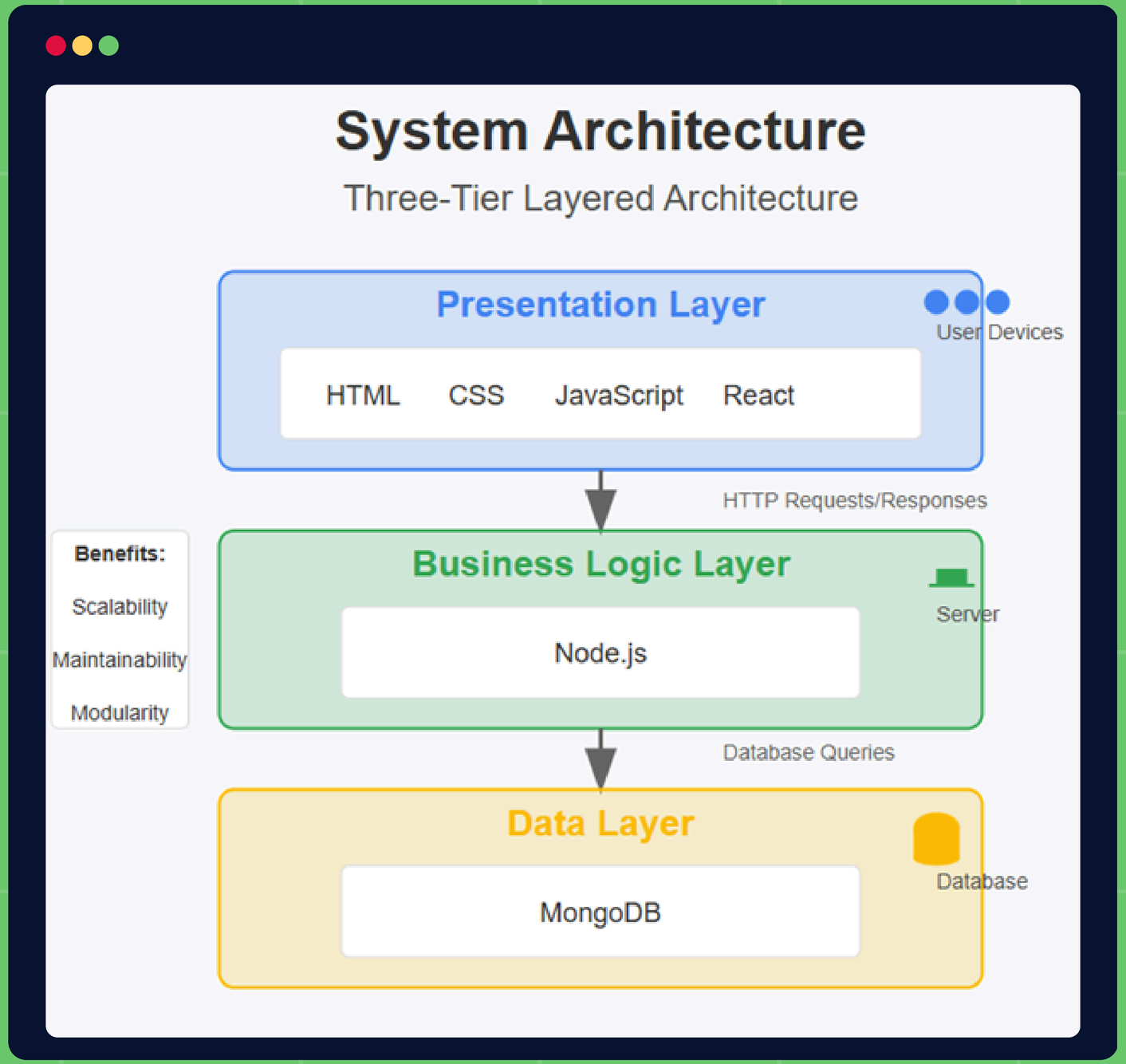
Areez Abdullah
Qasim Hassan
Muqarab Ahmad

Introduction

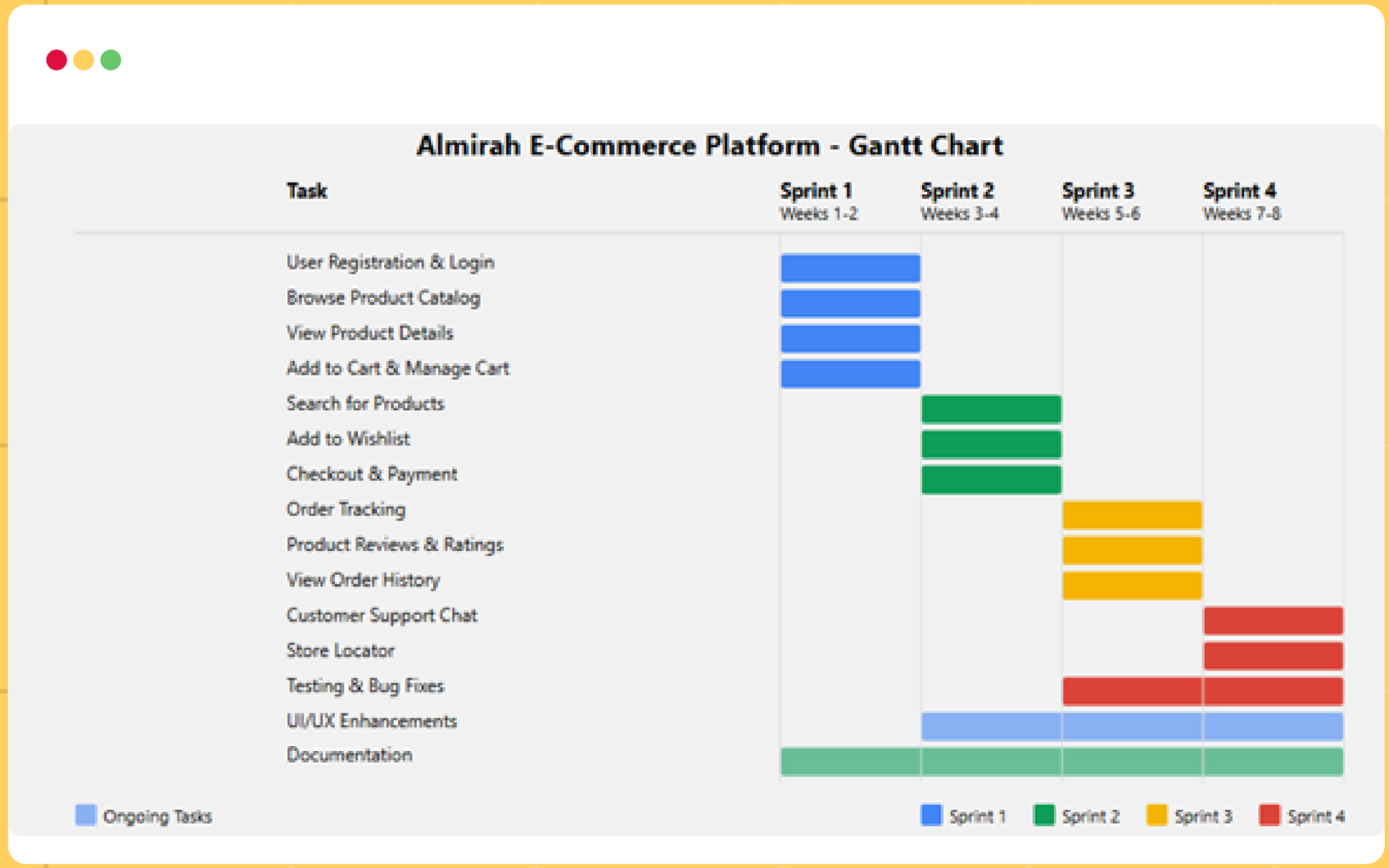
- The Almirah E-Commerce Platform modernizes online shopping for a renowned Pakistani fashion brand.
- Built using HTML, CSS, JavaScript, React, Node.js, and MongoDB.
- Focus: Seamless user experience, improved navigation, and accessibility.



System Architecture



Project Planner



- Followed Scrum methodology with 3 sprints.
- Key features implemented: User Registration, Product Browsing, Cart Management.
- Timeline: 8 weeks (Sprint 1: 2 weeks, Sprint 2: 2 weeks, Sprint 3: 2 weeks, Sprint 4: 2 weeks).

User Stories

- Total: 27 user stories.
- Sprint 1 Focus:
  - User Registration & Login
  - Browse & Search Products
  - View Product Details

Add to Cart & Manage Cart

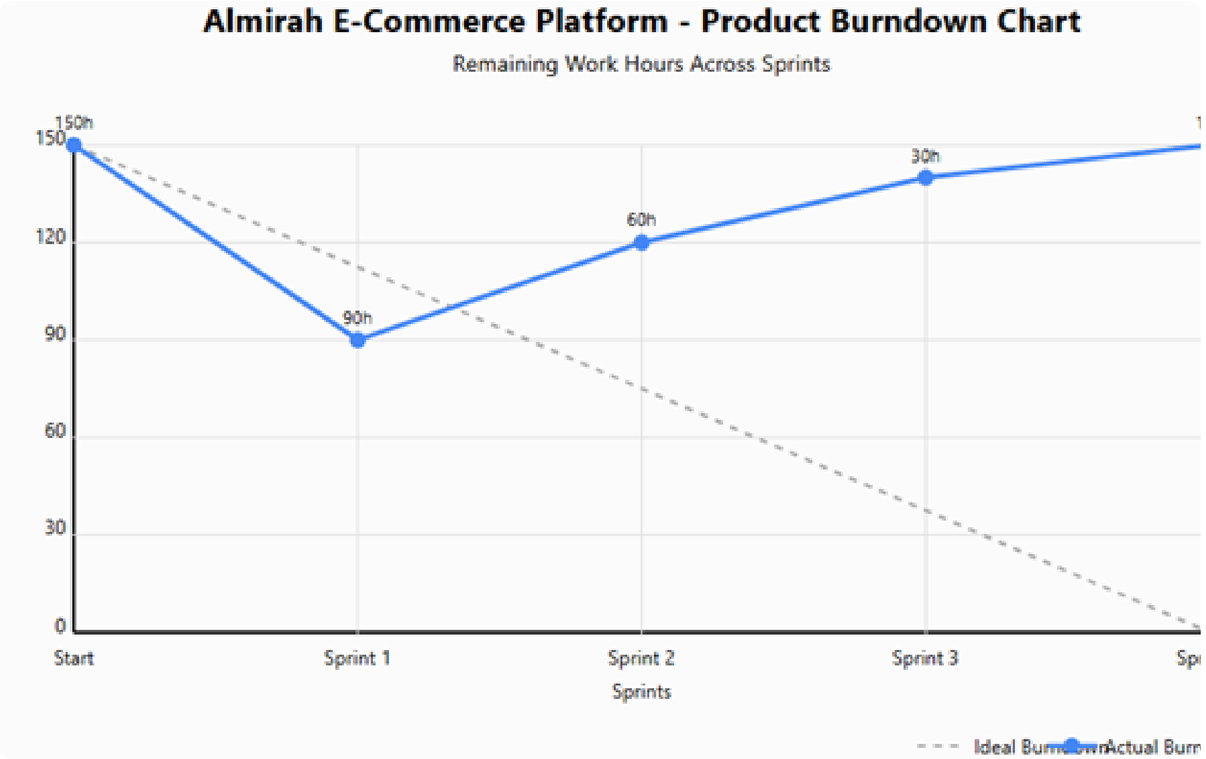
Sprint Backlog for Sprint 1 (Initial Development Phase)

We will prioritize **basic features** required for a functional website:

1. User Registration & Login
2. Browse & Search for Products
3. View Product Details
4. Add to Cart & Manage Cart

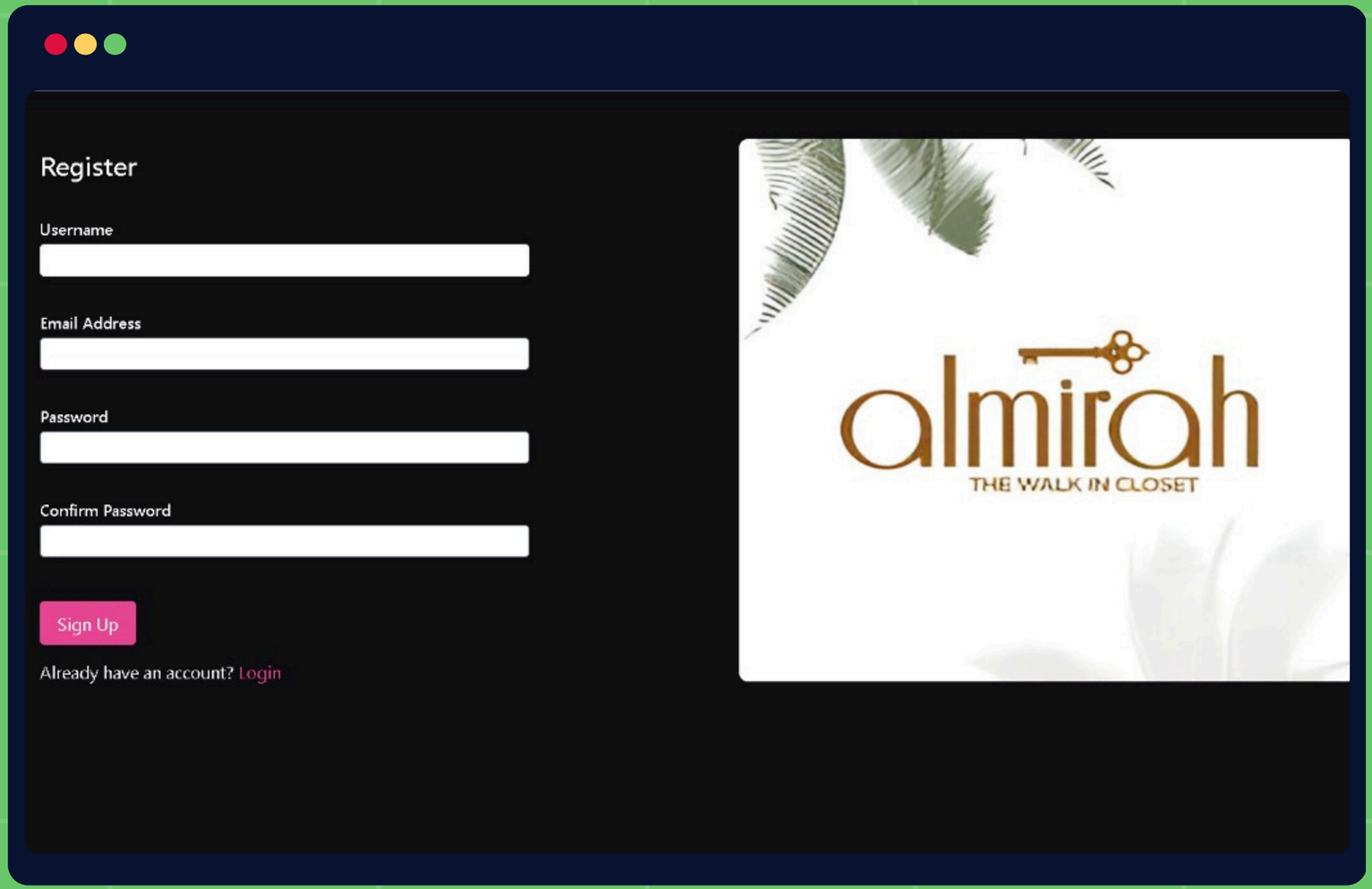
Burndown Chart

The burndown chart tracks the remaining work across sprints.



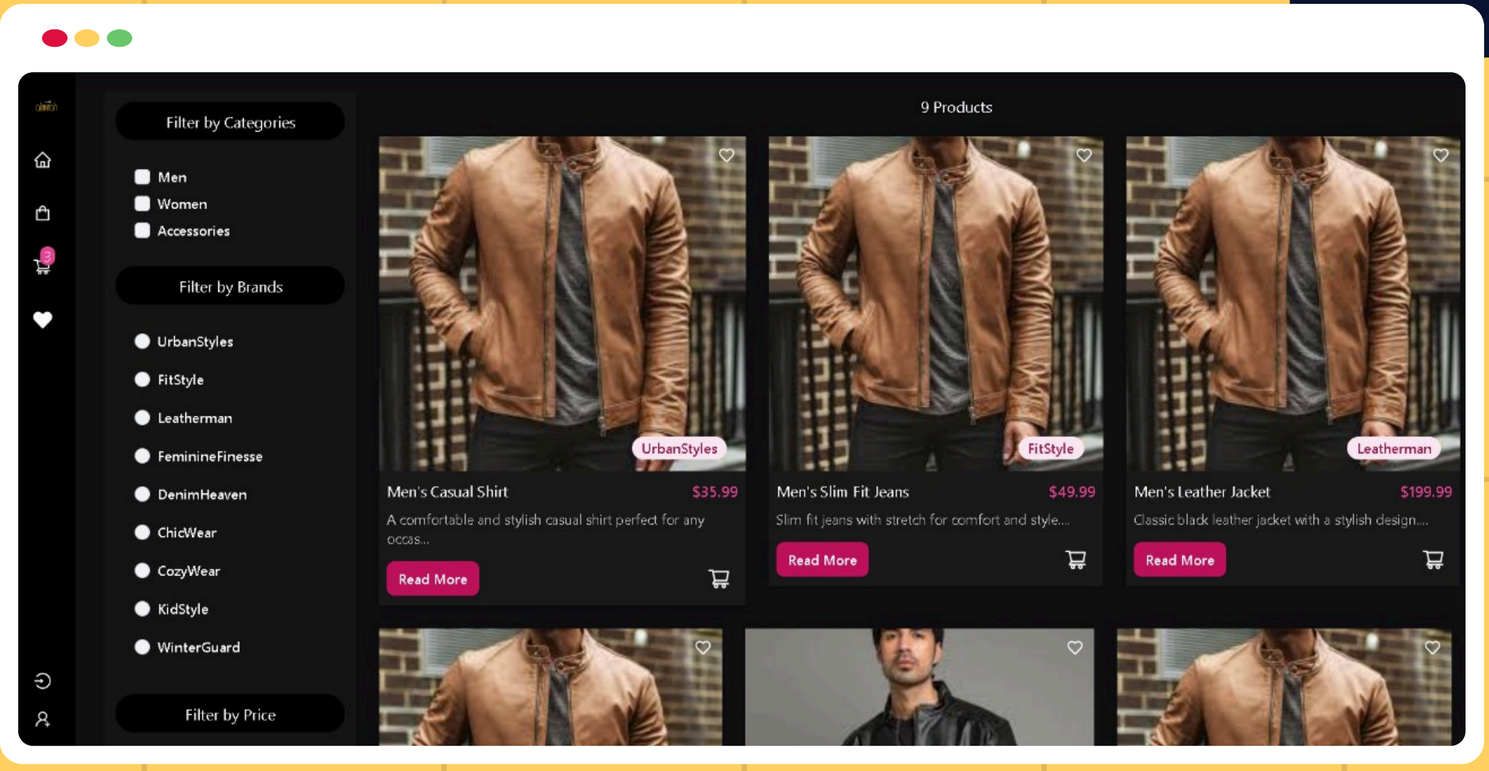
User Story Example - Registration

- As a new user, I want to sign up using my email and password so that I can create an account.
- Fields: Name, Email, Password
- Acceptance Criteria:
  - User can register with valid email and password.
  - Account is stored in the database.
  - Confirmation email sent.



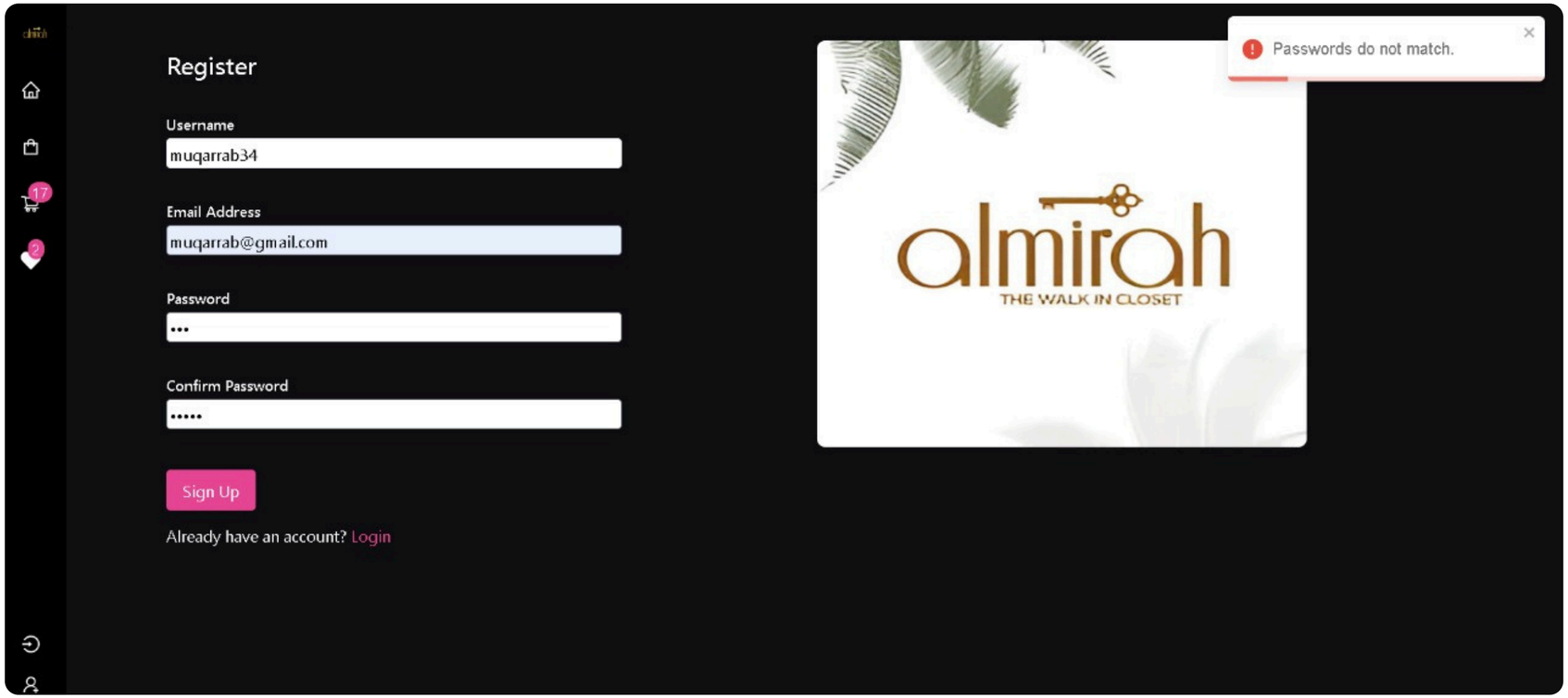
User Story Example - Product Browsing

- As a user, I want to browse products by categories so that I can easily find what I am looking for.
- Categories: Men’s Wear, Women’s Wear, Accessories
- Acceptance Criteria:
  - Products displayed by categories.
  - Categories are filterable and navigable.



Testing

- Used validation rules for input fields:
  - Name: Alphabetic, ≥3 chars
  - Email: Must include @ and .com
  - Password: 6-10 chars, with uppercase, lowercase, special char
  - Phone: 7-15 digits, numeric only
  - Ensures secure and accurate user input.



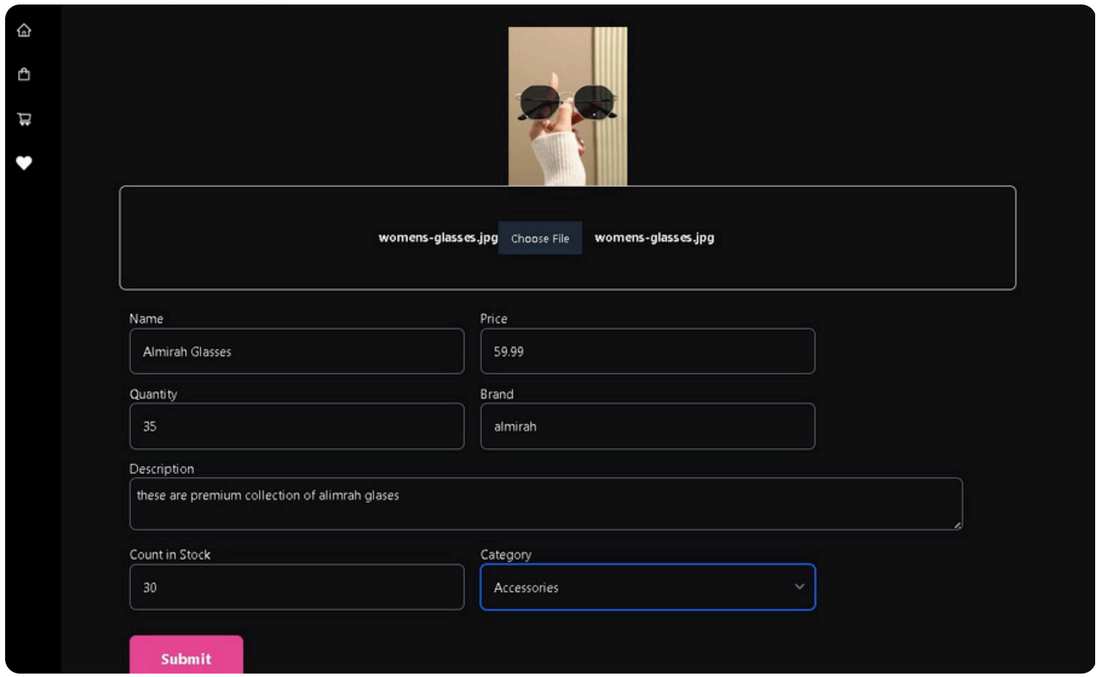
Black Box Testing

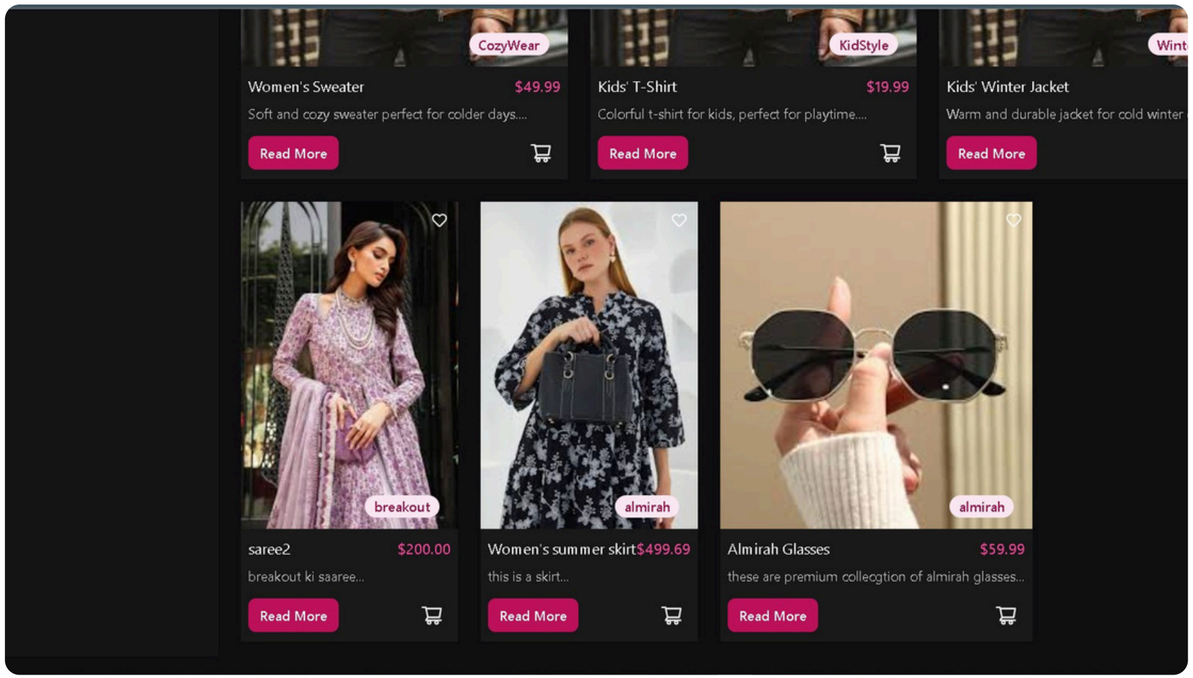
Test Case: Create a Product

- Test Objective: Verify that a product can be created successfully.
- Preconditions: Admin is logged in and on the "Create Product" page.
- Test Steps:

1. Go to the "Create Product" page.
2. Fill in all required fields (e.g., name, description, price, category, quantity).
3. Upload an image file for the product.
4. Click on the "Submit" button.

- Expected Result: The product is added to the system, a success message is displayed, and the product appears in the product list.





```
test('should reset fields when reset button is clicked', () => {
    const mockReset = jest.fn();
    render(<ProductForm onSubmit={jest.fn()} onReset={mockReset} />);
    fireEvent.click(screen.getByText('Reset'));
    expect(mockReset).toHaveBeenCalled();
  });
```

White Box Testing

```
[nodemon] restarting due to changes...
[nodemon] restarting due to changes...
[nodemon] starting `node index.js`
Server running on Port : 5000
```

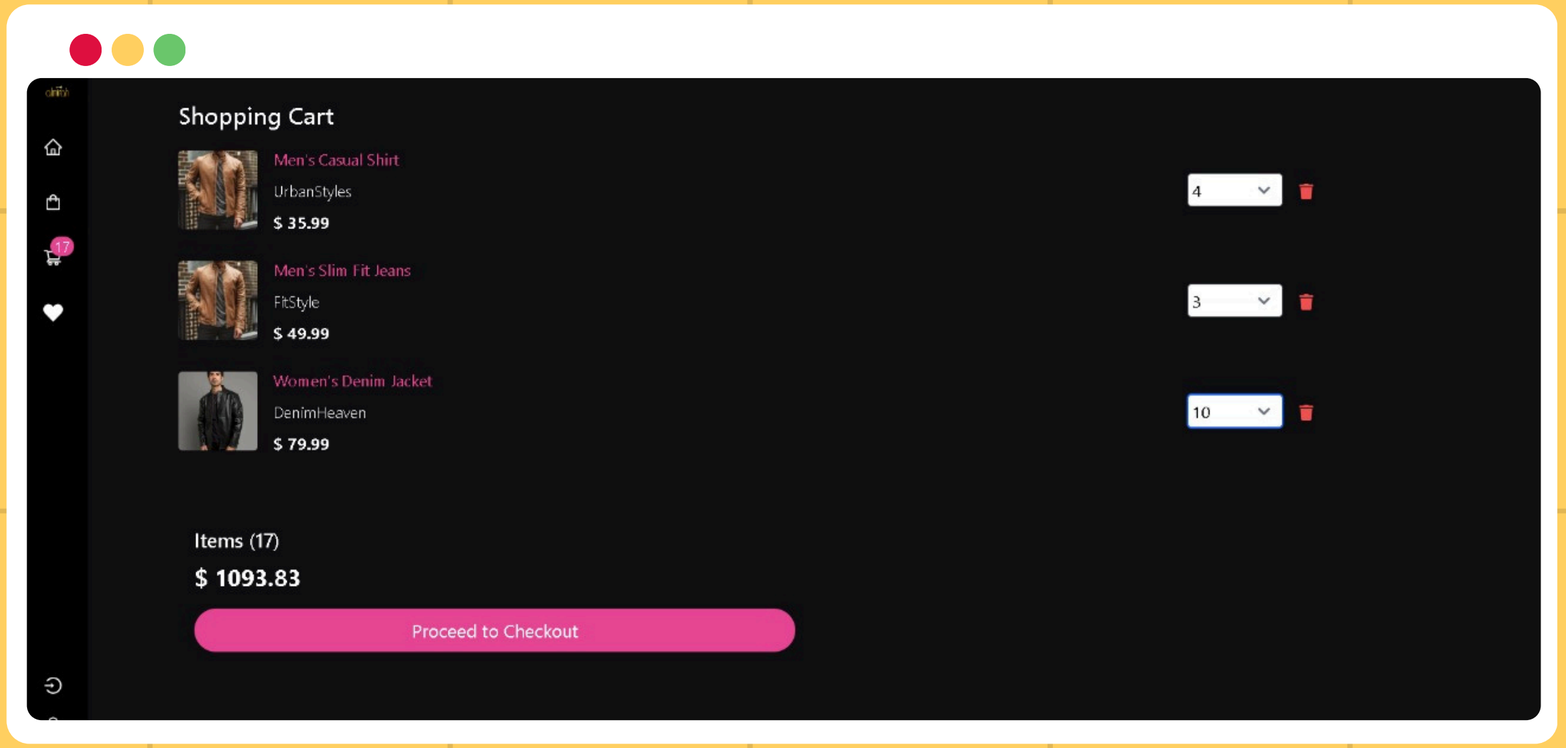
```
 RERUN  src/test/SmallProduct.test.jsx x2

 ❯ src/test/ProductList.test.jsx (0 test)

 Test Files 0 passed (1)
      Tests 0 passed (0)
   Start at 23:31:25
   Duration 4.07s
```

Conclusion

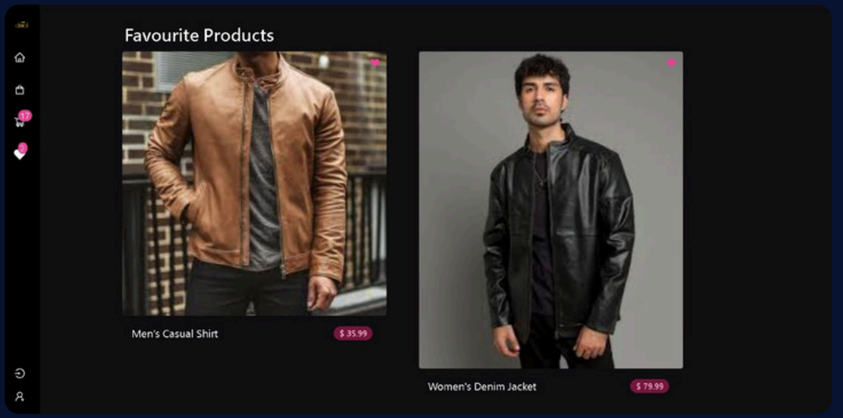
- Successfully delivered core features for Almirah’s online platform.
- Improved user experience with intuitive navigation and accessibility.
- Future sprints will add checkout, order tracking, and reviews.




Sign In
Email Address
Password
Sign In
New Customer? Register
almirah
THE WALK IN CLOSET



Thank You


Favourite Products
Men's Casual Shirt
$ 35.99
Women's Denim Jacket
$ 79.99